SVM Error: $\underbrace{\underset{(w^*, b^*)}{\arg\min} \frac{\|w\|}{2}}_{\text{Regularization}} + c \cdot \underbrace{\sum_{i=1}^{n} \tau_i}_{\text{hinge loss}}$

Kernel Trick. SVM

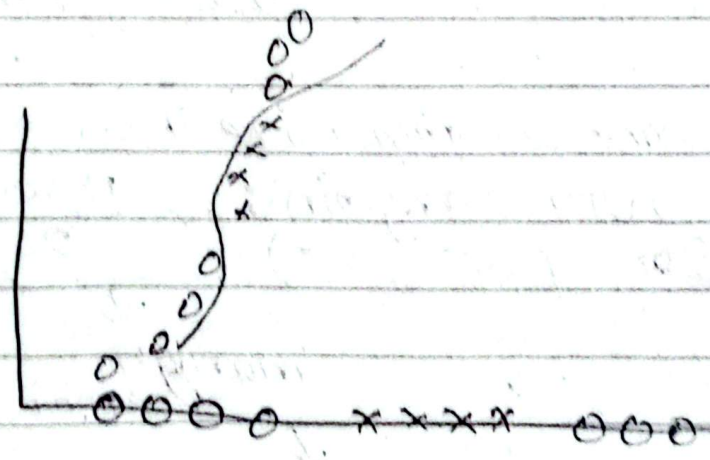
- In 1D: ——oooo——xxxx——ooo——

Here we can't separate by one point.

Hence, we use Kernel Trick. [1D - point, 2D - line, 3D - plane, nD - hyperplane]

Here we convert low dimension (i.e 1D) to higher dimension (i.e 2D) in such a way that they will be linear.

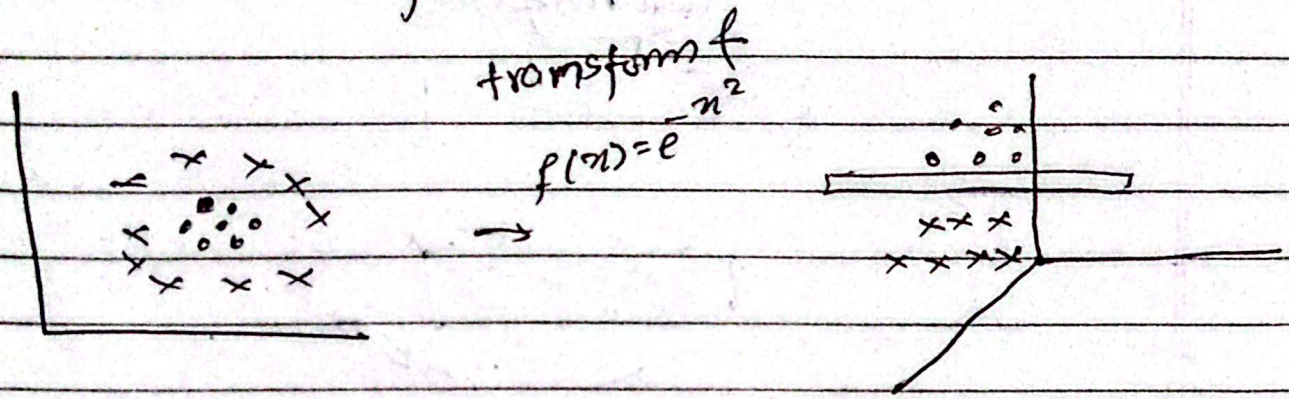
2D: x^2



$f(x) \rightarrow$ Kernel.

↳ types:
- RBF
- Polynomial
- Sigmoid.

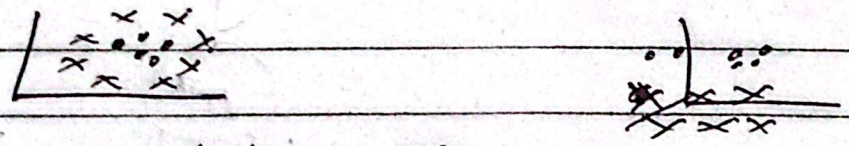
- The above transformation is known as Kernel transformation.



Code:

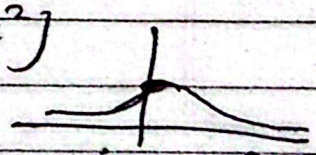
```
from sklearn.svm import SVC
clr = SVC(kernel = "linear")
clr.fit(X_train, y_train.ravel())
```

- Convert 2d to 3d



- use radial basis fn. $[y = e^{-x^2}]$

$[y = e^{-(x-s)^2}]$



central को आधार पर है।

- SVC (kernel = "rbf")

- SVC (kernel = "poly", degree = 3)

default also 3.